

Homework 3

Solution

1. Compute each of the following indefinite integrals:

(a) $\int e^{\sqrt{x}} dx$.

Solution:

We write

$$\int e^{\sqrt{x}} dx \stackrel{t=\sqrt{x} \rightarrow 2\sqrt{x}dt=dx}{=} \int e^t \cdot 2t dt = e^t \cdot 2t - \int e^t \cdot 2 dt = e^t \cdot 2t - e^t \cdot 2 + C = 2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C$$

(b) $\int \frac{dx}{\sin^2(x) \cos^2(x)}$.

Hint. use the formula $\sin^2(x) + \cos^2(x) = 1$ for every $x \in \mathbb{R}$.

Solution:

We write

$$\int \frac{dx}{\sin^2(x) \cos^2(x)} \stackrel{\text{formula}}{=} \int \frac{\sin^2(x) + \cos^2(x)}{\sin^2(x) \cos^2(x)} dx = \int \frac{dx}{\cos^2(x)} + \int \frac{dx}{\sin^2(x)} = \tan(x) - \cot(x) + C$$

(c) $\int x^2 \cdot \lambda e^{-\lambda x} dx$, where $\lambda \neq 0$.

Solution:

We develop

$$\begin{aligned} \int x^2 \cdot \lambda e^{-\lambda x} dx &\stackrel{\text{IBP}}{=} \lambda \left(-\frac{x^2 e^{-\lambda x}}{\lambda} - \int -\frac{2x e^{-\lambda x}}{\lambda} dx \right) = \lambda \left(-\frac{x^2 e^{-\lambda x}}{\lambda} + \frac{2}{\lambda} \int x e^{-\lambda x} dx \right) \stackrel{\text{IBP}}{=} \\ &\lambda \left[-\frac{x^2 e^{-\lambda x}}{\lambda} + \frac{2}{\lambda} \left(-\frac{x e^{-\lambda x}}{\lambda} - \int \frac{e^{-\lambda x}}{-\lambda} dx \right) \right] = \lambda \left[-\frac{x^2 e^{-\lambda x}}{\lambda} + \frac{2}{\lambda} \left(-\frac{x e^{-\lambda x}}{\lambda} + \frac{1}{\lambda} \int e^{-\lambda x} dx \right) \right] \\ &= \lambda \left[-\frac{x^2 e^{-\lambda x}}{\lambda} + \frac{2}{\lambda} \left(-\frac{x e^{-\lambda x}}{\lambda} + \frac{1}{\lambda} \left(\frac{e^{-\lambda x}}{-\lambda} \right) \right) \right] + C = \lambda \left(-\frac{x^2 e^{-\lambda x}}{\lambda} - \frac{2x e^{-\lambda x}}{\lambda^2} - \frac{2e^{-\lambda x}}{\lambda^3} \right) + C \\ &= -\frac{e^{-\lambda x} (\lambda^2 x^2 + 2x\lambda + 2)}{\lambda^2} + C \end{aligned}$$

(d) $\int \frac{\sqrt{x}}{x+1} dx.$

Solution:

We develop

$$\begin{aligned} \int \frac{\sqrt{x}}{x+1} dx &\stackrel{t=\sqrt{x} \rightarrow dt=\frac{1}{2\sqrt{x}} dx}{=} \int \frac{t}{t^2+1} 2t dt = 2 \int \frac{t^2}{t^2+1} dt = 2 \int \frac{t^2+1-1}{t^2+1} dt = 2 \int \left(1 - \frac{1}{t^2+1}\right) dt \\ &= 2(t - \arctan t) + C = 2(\sqrt{x} - \arctan(\sqrt{x})) + C \end{aligned}$$

(e) $\int \frac{dx}{\sqrt{x} + \sqrt[3]{x}}.$

Solution:

Set $t = \sqrt[6]{x}$. Then $dt = \frac{1}{6}x^{-\frac{5}{6}}dx = \frac{1}{6}t^{-5}$, so $dx = 6t^5 dt$. It follows that

$$\int \frac{dx}{\sqrt{x} + \sqrt[3]{x}} = \int \frac{6t^5 dt}{t^3 + t^2} = \int \frac{6t^5}{t^2(t+1)} dt = \int \frac{6t^3}{t+1} dt.$$

Now perform polynomial long division:

$$\frac{t^3}{t+1} = t^2 - t + 1 - \frac{1}{t+1}.$$

Hence

$$\int \frac{6t^3}{t+1} dt = 6 \int \left(t^2 - t + 1 - \frac{1}{t+1}\right) dt = 6 \left(\frac{t^3}{3} - \frac{t^2}{2} + t - \ln|t+1|\right) + C = 2t^3 - 3t^2 + 6t - 6 \ln|t+1| + C.$$

Substituting back $t = x^{1/6}$, we obtain

$$\int \frac{dx}{\sqrt{x} + \sqrt[3]{x}} = 2\sqrt{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6 \ln(\sqrt[6]{x} + 1) + C.$$

2. Compute each of the following indefinite integrals:

(a) $\int \sin(\ln x) \, dx$.

Solution:

We develop:

$$\begin{aligned} \int \sin(\ln x) \, dx &\stackrel{\text{IBP}}{=} x \cdot \sin(\ln x) - \int x \cdot \cos(\ln x) \cdot \frac{1}{x} \, dx \\ &\stackrel{\text{IBP}}{=} x \cdot \sin(\ln x) - \left(x \cdot \cos(\ln x) + \int x \cdot \sin(\ln x) \cdot \frac{1}{x} \, dx \right) \end{aligned}$$

Rearranging, we obtain that

$$2 \int \sin(\ln x) \, dx = x \cdot \sin(\ln x) - x \cdot \cos(\ln x) + C$$

Dividing both sides of the equation by 2, we conclude that

$$\int \sin(\ln x) \, dx = \frac{x \cdot \sin(\ln x) - x \cdot \cos(\ln x)}{2} + C$$

(b) $\int \arctan(\sqrt{x}) \, dx$.

Solution:

We write

$$\begin{aligned} \int \arctan(\sqrt{x}) \, dx &\stackrel{\text{IBP}}{=} x \arctan(\sqrt{x}) - \int \left[x \cdot \frac{\frac{1}{2\sqrt{x}}}{1+x} \right] dx \\ &= x \arctan(\sqrt{x}) - \int \frac{\sqrt{x}}{2(1+x)} dx \\ &\stackrel{t=\sqrt{x}, dt=\frac{1}{2\sqrt{x}} dx=\frac{1}{2t} dx}{=} x \arctan(\sqrt{x}) - \frac{1}{2} \int \frac{t}{1+t^2} 2t dt \\ &= x \arctan(\sqrt{x}) - \int \frac{t^2+1-1}{1+t^2} dt = x \arctan(\sqrt{x}) - (t - \arctan t) + C \\ &= x \arctan(\sqrt{x}) - \sqrt{x} + \arctan(\sqrt{x}) + C = \arctan(\sqrt{x})(x+1) - \sqrt{x} + C \end{aligned}$$

3. (a) Compute the indefinite integral $\int \frac{dx}{\cos(x)}$ by writing

$$\frac{1}{\cos(x)} = \frac{\cos(x)}{\cos^2(x)} = \frac{\cos(x)}{1 - \sin^2(x)} = \frac{\cos(x)}{(1 - \sin(x))(1 + \sin(x))} = \frac{1}{2} \cdot \frac{\cos(x)}{1 + \sin(x)} + \frac{1}{2} \cdot \frac{\cos(x)}{1 - \sin(x)}.$$

Solution:

We write

$$\begin{aligned} \int \frac{dx}{\cos(x)} &= \frac{1}{2} \int \frac{\cos(x)}{1 + \sin(x)} dx + \frac{1}{2} \int \frac{\cos(x)}{1 - \sin(x)} dx = \frac{1}{2} \ln |1 + \sin(x)| - \frac{1}{2} \ln |1 - \sin(x)| + C \\ &= \frac{1}{2} \ln \left| \frac{1 + \sin(x)}{1 - \sin(x)} \right| + C \end{aligned}$$

(b) Compute $\int \frac{dx}{\cos(x)}$ by making the substitution $t = \tan\left(\frac{x}{2}\right)$. Note that $\frac{2}{1-t^2} = \frac{1}{1+t} + \frac{1}{1-t}$.

Solution:

We make the substitution $t = \tan\left(\frac{x}{2}\right)$. Then

$$dt = \frac{\frac{1}{2}dx}{\cos^2\left(\frac{x}{2}\right)} \stackrel{1+t^2 = \frac{1}{\cos^2\left(\frac{x}{2}\right)}}{=} \frac{1}{2}(1+t^2)dx \Rightarrow dx = \frac{2}{1+t^2}dt$$

We also know that for every $x \in \mathbb{R}$

$$\cos(x) = 2 \cos^2\left(\frac{x}{2}\right) - 1 \stackrel{1+t^2 = \frac{1}{\cos^2\left(\frac{x}{2}\right)}}{=} \frac{2}{1+t^2} - 1 = \frac{1-t^2}{1+t^2}$$

It follows that

$$\begin{aligned} \int \frac{dx}{\cos(x)} &= \int \frac{1+t^2}{1-t^2} \cdot \frac{2}{1+t^2} dt = \int \frac{2dt}{1-t^2} = \int \frac{dt}{1+t} + \int \frac{dt}{1-t} = \ln|1+t| - \ln|1-t| + C \\ &= \ln\left|1 + \tan\left(\frac{x}{2}\right)\right| - \ln\left|1 - \tan\left(\frac{x}{2}\right)\right| + C = \ln\left|\frac{1 + \tan\left(\frac{x}{2}\right)}{1 - \tan\left(\frac{x}{2}\right)}\right| + C \end{aligned}$$

(c) Conclude that for every $-\frac{\pi}{2} < x < \frac{\pi}{2}$ we have

$$\frac{1 + \sin(x)}{1 - \sin(x)} = \left(\frac{1 + \tan\left(\frac{x}{2}\right)}{1 - \tan\left(\frac{x}{2}\right)}\right)^2.$$

Solution:

As two anti-derivatives of the same function differ by a constant on an interval, there exists a $C \in \mathbb{R}$ such that for every $-\frac{\pi}{2} < x < \frac{\pi}{2}$ we have

$$\frac{1}{2} \ln\left|\frac{1 + \sin(x)}{1 - \sin(x)}\right| = \ln\left|\frac{1 + \tan\left(\frac{x}{2}\right)}{1 - \tan\left(\frac{x}{2}\right)}\right| + C.$$

Plugging $x = 0$ we obtain $C = 0$ and therefore for every $-\frac{\pi}{2} < x < \frac{\pi}{2}$ we have

$$\ln\sqrt{\left|\frac{1 + \sin(x)}{1 - \sin(x)}\right|} = \frac{1}{2} \ln\left|\frac{1 + \sin(x)}{1 - \sin(x)}\right| = \ln\left|\frac{1 + \tan\left(\frac{x}{2}\right)}{1 - \tan\left(\frac{x}{2}\right)}\right|$$

and therefore $\left|\frac{1 + \sin(x)}{1 - \sin(x)}\right| = \left|\frac{1 + \tan\left(\frac{x}{2}\right)}{1 - \tan\left(\frac{x}{2}\right)}\right|^2 = \left(\frac{1 + \tan\left(\frac{x}{2}\right)}{1 - \tan\left(\frac{x}{2}\right)}\right)^2$.

Finally, since $-1 \leq \sin(x) < 1$ for every $-\frac{\pi}{2} < x < \frac{\pi}{2}$, the left-hand side is non-negative, and therefore for every $-\frac{\pi}{2} < x < \frac{\pi}{2}$ we have

$$\frac{1 + \sin(x)}{1 - \sin(x)} = \left(\frac{1 + \tan\left(\frac{x}{2}\right)}{1 - \tan\left(\frac{x}{2}\right)}\right)^2.$$

4. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function. Suppose that

$$\int \ln(x) f'(x) dx = (\ln(x) - 1)f(x) + C.$$

Prove that there exists an $\alpha \in \mathbb{R}$ such that

$$f(x) = \alpha x, \quad \forall x > 0.$$

Solution:

Consider the function defined by

$$g(x) = \frac{f(x)}{x}, \quad \forall x \in (0, \infty)$$

Note that g is differentiable by AOD and

$$g'(x) = \frac{xf'(x) - f(x)}{x^2}, \quad \forall x \in (0, \infty)$$

By the assumption, the function $(\ln(x) - 1)f(x)$ is an anti-derivative of the function $\ln(x)f'(x)$ on the interval $(0, \infty)$, namely

$$[(\ln(x) - 1)f(x)]' = \ln(x)f'(x), \quad \forall x \in (0, \infty)$$

On the other hand, by Leibniz rule

$$[(\ln(x) - 1)f(x)]' = \frac{f(x)}{x} + (\ln(x) - 1)f'(x), \quad \forall x \in (0, \infty)$$

Equating, we obtain

$$\ln(x)f'(x) = \frac{f(x)}{x} + (\ln(x) - 1)f'(x) \Rightarrow xf'(x) - f(x) = 0, \quad \forall x \in (0, \infty)$$

Hence

$$g'(x) = \frac{xf'(x) - f(x)}{x^2} = 0, \quad \forall x \in (0, \infty)$$

It follows that there exists an $\alpha \in \mathbb{R}$ such that for every $x \in (0, \infty)$,

$$g(x) = \alpha \Rightarrow \frac{f(x)}{x} = \alpha \Rightarrow f(x) = \alpha x$$

5. (a) Student A has proved that $0 = 1$ by using the method of integration by parts in the following way:

$$\int \frac{dx}{x \cdot \ln(x)} \stackrel{u'=\frac{1}{x}, v=\frac{1}{\ln(x)}}{=} \frac{1}{\ln(x)} \cdot \ln(x) - \int \left[\ln(x) \cdot \left(-\frac{1}{x \cdot \ln^2(x)} \right) \right] dx = 1 + \int \frac{dx}{x \cdot \ln(x)}$$

subtracting $\int \frac{dx}{x \cdot \ln(x)}$ from both sides, we get $0 = 1$.

Explain Student A's mistake.

Solution:

The value of an indefinite integral is not a particular value (i.e., not a number) but rather a range of values which differ by a constant. Thus, one should understand the equality above as equality up to a constant. It follows that the law of cancellation does not apply in his proof.

- (b) Student A proved that every bounded function is integrable on every interval $[a, b]$ in the following way:

Let $a, b \in \mathbb{R}$ such that $a < b$. As f is bounded on $[a, b]$, then there exists a $K > 0$ such that $|f(x)| \leq K$ for every $x \in [a, b]$. For every n we choose a partition $P_n = (x_0, \dots, x_n)$ of $[a, b]$ for which $x_{k+1} - x_k < \boxed{\frac{1}{K \cdot n^2}}$ for every $0 \leq k \leq n-1$. Thus

$$\begin{aligned} 0 \leq \bar{S}(f, P_n) - \underline{S}(f, P_n) &= \sum_{k=1}^n M_k (x_{k+1} - x_k) - \sum_{k=1}^n m_k (x_{k+1} - x_k) = \sum_{k=1}^n (M_k - m_k) (x_{k+1} - x_k) < \sum_{k=1}^n (M_k - m_k) \frac{1}{K \cdot n^2} \\ &\leq \sum_{k=1}^n 2K \frac{1}{K \cdot n^2} = 2 \sum_{k=1}^n \frac{1}{n^2} = \frac{2}{n} \end{aligned}$$

Thus, by the squeeze rule, we have $\lim_{n \rightarrow \infty} [\bar{S}(f, P_n) - \underline{S}(f, P_n)] = 0$, from which follows that f is integrable.

Explain student A's mistake.

Solution:

Student A is wrong, since a partition P_n satisfying the requirements does not necessarily exist. Indeed, for the Dirichlet function D and the interval $[0, 1]$, there is no partition of size $n = 3$ satisfying $x_{k+1} - x_k < \frac{1}{9}$ for every $0 \leq k \leq 2$.

6. Let $a, b \in \mathbb{R}$ such that $a < b$ and let f be a function that is bounded on $[a, b]$.

(a) Suppose that there exists a sequence of partitions $(P_n)_{n=1}^{\infty}$ of $[a, b]$ for which

$$\lim_{n \rightarrow \infty} [\bar{S}(f, P_n) - \underline{S}(f, P_n)] = 0$$

Prove that f is integrable on $[a, b]$ and that

$$\lim_{n \rightarrow \infty} \bar{S}(f, P_n) = \lim_{n \rightarrow \infty} \underline{S}(f, P_n) = \int_a^b f(x) dx$$

Solution:

We know that for every $n \in \mathbb{N}$ we have

$$\underline{S}(f, P_n) \leq \underline{I}(f) \leq \bar{I}(f) \leq \bar{S}(f, P_n)$$

Since $\lim_{n \rightarrow \infty} [\bar{S}(f, P_n) - \underline{S}(f, P_n)] = 0$, and as $\lim_{n \rightarrow \infty} \underline{I}(f) = \underline{I}(f)$, then by the inverse squeeze rule for sequences we have

$$\lim_{n \rightarrow \infty} \bar{S}(f, P_n) = \lim_{n \rightarrow \infty} \underline{S}(f, P_n) = \underline{I}(f)$$

Similarly,

$$\lim_{n \rightarrow \infty} \bar{S}(f, P_n) = \lim_{n \rightarrow \infty} \underline{S}(f, P_n) = \bar{I}(f)$$

It follows that $\underline{I}(f) = \bar{I}(f)$, and therefore f is integrable and

$$\lim_{n \rightarrow \infty} \bar{S}(f, P_n) = \lim_{n \rightarrow \infty} \underline{S}(f, P_n) = \bar{I}(f) = \underline{I}(f) = \int_a^b f(x) dx$$

(b) Suppose that f is *Lipschitz*, namely,

$$\exists M > 0 : \forall x, y \in [a, b] \text{ we have } |f(x) - f(y)| \leq M \cdot |x - y|$$

i. Prove that for every $a \leq x_1 < x_2 \leq b$ we have

$$\sup \{f(x) : x_1 \leq x \leq x_2\} - \inf \{f(x) : x_1 \leq x \leq x_2\} \leq M \cdot (x_2 - x_1)$$

Solution:

Let $x_1, x_2 \in [a, b]$ such that $x_1 < x_2$. By the assumption, we know that for every $x, y \in [x_1, x_2]$ it holds that

$$f(x) - f(y) \leq |f(x) - f(y)| \leq M \cdot |x - y| \leq M \cdot (x_2 - x_1)$$

Since we know that for bounded sets $\emptyset \neq A, B \subseteq \mathbb{R}$ we have $\sup(A - B) = \sup(A) - \inf(B)$, then

$$\begin{aligned} & \sup \{f(x) : x_1 \leq x \leq x_2\} - \inf \{f(y) : x_1 \leq y \leq x_2\} \\ &= \sup \{f(x) - f(y) : x_1 \leq x, y \leq x_2\} \stackrel{\text{minimality}}{\leq} M \cdot (x_2 - x_1) \end{aligned}$$

ii. Given a partition of size n , $P = (x_0, \dots, x_n)$ of $[a, b]$, we define

$$\Delta(P) = \max \{x_k - x_{k-1} : 1 \leq k \leq n\}$$

Let $(P_n)_{n=1}^\infty$ be a sequence of partitions of $[a, b]$ for which P_n is a partition of size n for every $n \in \mathbb{N}$ and such that

$$\lim_{n \rightarrow \infty} \Delta(P_n) = 0$$

Prove that

$$\lim_{n \rightarrow \infty} [\bar{S}(f, P_n) - \underline{S}(f, P_n)] = 0$$

and conclude that f is integrable on $[a, b]$ and

$$\lim_{n \rightarrow \infty} \bar{S}(f, P_n) = \lim_{n \rightarrow \infty} \underline{S}(f, P_n) = \int_a^b f(x) dx$$

Solution:

By the previous item, for every n and for every $1 \leq k \leq n$, it holds that

$$M_k - m_k = \sup \{f(x) : x_{k-1} \leq x \leq x_k\} - \inf \{f(y) : x_{k-1} \leq y \leq x_k\} \leq M \cdot (x_k - x_{k-1}) \leq M \cdot \Delta(P_n)$$

Thus,

$$\begin{aligned} 0 \leq \bar{S}(f, P_n) - \underline{S}(f, P_n) &= \sum_{k=1}^n M_k \cdot (x_k - x_{k-1}) - \sum_{k=1}^n m_k \cdot (x_k - x_{k-1}) = \sum_{k=1}^n (M_k - m_k) \cdot (x_k - x_{k-1}) \\ &\leq M \cdot \Delta(P_n) \cdot \sum_{k=1}^n (x_k - x_{k-1}) = M \cdot \Delta(P_n) \cdot (b - a) \end{aligned}$$

Taking the limit as $n \rightarrow \infty$, we obtain that $\lim_{n \rightarrow \infty} [\bar{S}(f, P_n) - \underline{S}(f, P_n)] = 0$. Thus, by item (i), it follows that f is integrable and

$$\lim_{n \rightarrow \infty} \bar{S}(f, P_n) = \lim_{n \rightarrow \infty} \underline{S}(f, P_n) = \int_a^b f(x) dx$$